

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Business development Trainee”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning



Intellipaath

Submitted by:

ANUKALP JAIN - 1DS20AI009

Conducted at

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Business Development Trainee**” carried out by **Mr Anukalp Jain** bearing USN **1DS20AI009**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)



Signature of Reporting Head
Bhargavi Narayana A S
Associate Vice President

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Anukalp Jain**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **Intellipaath**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

NAME: ANUKALP JAIN

Place: Bangalore

USN: 1DS20AI009

OFFER LETTER



6th Floor, Primeco Towers,
Arekere Gate Junction,
Bannerghatta Main Road,
Bangalore -560076
(M):+917022374614.
www.intellipaat.com

To

Date: 20th February, 2024

Anukalp Jain

Subject: Internship Offer Letter

Dear Anukalp Jain

In reference to your application, we would like to congratulate you on being selected for an internship with the IntelliPaat Software Solutions Pvt. Ltd. Your work is scheduled to start on **26th February, 2024** for a period of **6** months. During this period, you will get paid **Rs. 22,000/month** (Rupees Twenty-Two Thousand Only) and you will be working as a '**Business Development Trainee**'. The technical platform and job role will be shared with you by your manager on the day of your joining.

Note:

- Number of working days is **6** days a week.
- Your First month stipend(30days)will be on hold and it will be released along with your fourth month stipend.
- As per your KPI, you will be expected to have a minimum of 3 hours of call time. Failing to meet your KPI, will automatically fall into absenteeism
- During this Internship, you are eligible to get up to **Rs 200,000/-** as an incentive based on your performance.
- During your internship, you are expected to be disciplined and sincere towards your job responsibility.
- Based on your performance, the Pre-Placement Offer will be released before the completion of your internship at IntelliPaat.
- The company reserves the right to pay or recover salary in lieu of the notice period (if applicable) or to relieve you before the expiry of the notice period. All payments and recoveries made under this clause will be based on Gross Salary
- You need to serve **30** days of notice period without fail, or else the management of IntelliPaat can hold your salary/ experience letter / relieving letter/ any background verification in the future.
- Minimum working duration must be 9 hours including a one-hour lunch/dinner break.
- After the successful completion of 6 Months Internship period total CTC would be **Rs 7,25,000** INR.
- No examination preparatory leaves will be provided during this period
- If the college/university exams are pending then you are allowed to take leaves for exams and for applying for the same you need to get the letter from your TPO with examination dates and get it approved by your manager.
- No leaves would be allowed during 1st month of your employment (allowed only if it is genuine). If you are takingleaves we would be requiring supporting and admissible documents for the same
- Failing to do the same organization can be liable to take action against your employment

Again, congratulations and we look forward to working with you.

Yours truly,

For IntelliPaat Software Solutions Private Limited



Bhargavi Narayan A S

Associate Vice President- Human Resources

IntelliPaat Software Solutions Pvt. Ltd.

6th Floor, Primeco Towers, Arekere Gate Junction, Bannerghatta Main Road, Bengaluru,
Karnataka-560076

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mrs Bhargavi Narayan A S, Intellipaat, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ANUKALP JAIN
1DS20AI009

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CHAPTER 1

INTRODUCTION

Intellipaat stands as a beacon in the realm of online professional training, a global entity committed to shaping the careers of professionals worldwide. With an exhaustive array of certification programs spanning over 150 technologies, Intellipaat remains at the forefront of innovation and industry relevance.

Our mission is clear: to empower professionals with the latest knowledge and skills demanded by today's dynamic job market. Whether it's mastering Big Data, delving into the intricacies of Data Science & AI, navigating the complexities of Business Intelligence, or exploring the limitless possibilities of Cloud and Blockchain technologies, Intellipaat provides the stepping stones for success.

What sets Intellipaat apart is our unwavering dedication to quality and excellence. Our trainers, possessing over a decade of industry experience, bring unparalleled expertise to the virtual classroom. Through a combination of comprehensive theoretical knowledge and extensive hands-on projects, learners are equipped not only with theoretical understanding but also with practical skills vital for real-world application.

At Intellipaat, we understand the importance of continuous learning and professional growth. We meticulously evaluate learner progress, ensuring that each individual achieves their maximum potential. Our industry-recognized certifications serve as badges of honor, validating the expertise and proficiency gained through our programs.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

DailyPe Intellipaat is a leading global provider of online professional training, specializing in a wide range of certification programs covering over 150 technologies. Established with a commitment to empowering professionals with the latest industry-relevant skills, Intellipaat offers comprehensive training in areas such as Big Data, Data Science & AI, Business Intelligence, Cloud Computing, Blockchain, Database Management, Programming, Testing, SAP, and many more.

What sets Intellipaat apart is its dedication to quality and excellence. The company's trainers bring over a decade of industry experience to the table, ensuring that learners receive top-notch instruction and guidance. Through a blend of theoretical knowledge and hands-on projects, Intellipaat equips learners with both the understanding and practical skills necessary to excel in their chosen fields.

One of Intellipaat's key strengths lies in its rigorous evaluation process, ensuring that learners make tangible progress and achieve their learning objectives. Upon completion of their programs, learners receive industry-recognized certifications, validating their expertise and enhancing their career prospects.

In addition to serving individual learners, Intellipaat also collaborates with corporate clients to upskill their workforce. By tailoring training solutions to meet the specific needs of organizations, Intellipaat helps businesses stay competitive in today's fast-paced digital landscape.

CHAPTER 3

SCOPE OF WORK

Test Planning

Developing test strategies and plans that outline the testing objectives, resources, timelines, and methodologies. Determining the scope of testing activities and the resources required, including personnel and testing tools.

Test Design

Designing test cases, test scripts, and test data based on software requirements and specifications. Creating traceability matrices to ensure comprehensive test coverage that maps test cases to their respective requirements.

Test Execution

Conducting various types of testing such as functional testing, integration testing, system testing, and user acceptance testing. Executing test cases manually or using automated test tools. Recording test results and documenting any defects or issues found during testing

Defect Management

Conducting various types of testing such as functional testing, integration testing, system testing, and user acceptance testing. Executing test cases manually or using automated test tools. Recording test results and documenting any defects or issues found during testing.

Test Automation

Developing and maintaining automated test scripts to improve testing efficiency and coverage. Selecting appropriate automation tools and frameworks that align with the technology stack of the software being tested.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

Understanding Testing Fundamentals:

Learn the basic principles and concepts of software testing, including the software development lifecycle (SDLC) and the role of testing within it. Gain familiarity with different types of testing (e.g., unit, integration, system, acceptance)

Mastering Test Design Techniques:

Develop the ability to design comprehensive test cases that cover all aspects of application functionality. Learn to use different test design techniques such as boundary value analysis, equivalence partitioning, and state transition testing.

Proficiency in Test Execution and Management:

Acquire skills in executing test cases effectively and managing test cycles. Learn how to document test results accurately and manage defects from discovery to resolution.

Expertise in Automated Testing:

Learn to develop and maintain automated test scripts using popular automation tools and frameworks. Understand best practices in selecting test cases for automation and maintaining test scripts.

Developing Soft Skills::

Enhance communication skills to effectively interact with team members, developers, and stakeholders. Improve problem-solving skills and the ability to think critically to identify potential issues during the testing phases.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout my internship at Intellipaat, I encountered challenges that are common experiences for developers, especially when working on a complex tasks.

1. Diversity in Testing Environments

Apps and web applications are accessed from thousands of device-browser-platform combinations simultaneously. Needless to say, teams need to develop robust applications that run flawlessly across the most widely used combinations. However, given the increasing number of mobile devices already present in the market (plus new ones being released), it is challenging to test applications on every latest mobile device rolled out.

Solution

An effective solution is to opt for a cloud-based testing platform like BrowserStack. Its real device cloud empowers teams to choose from 2000+ real devices and browsers for testing apps and websites on multiple device-browser-OS combinations. .

2. Inadequate Testing

For an application to be robust and flawless, it needs to be thoroughly tested in different environments. Specifically for complex applications, testers need to ensure that every single bit of code undergoes regression tests. This takes considerable time. However, clients expect software teams to deliver products in shorter timelines so as to maintain their edge over competitors. This creates pressure on the testing teams to pace up testing. In some cases, QAs also miss out on prioritizing key test cases which leads to insufficient test coverage. This leads to inadequate testing and lets bugs escape into production.

Solution

Teams must plan and prioritize their test cases. One must also evaluate which tests require automation and which tests need to be executed manually. Finding the right balance between manual and automated testing is the key to successful testing.

Planning and prioritizing tests will help teams evaluate critical areas that need more attention in the earlier stages and save time later on. This approach will also ensure that critical areas of the software are tested thoroughly in time

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Development of Automated Testing Tools

Contribution: Automated testing tools have revolutionized the way testing is conducted by enabling rapid execution of test cases, ensuring consistency, and improving efficiency.

Achievement: Tools like Selenium, Appium, and Jenkins have become integral in continuous integration and continuous delivery pipelines, significantly reducing the time to market for software products.

Agile and DevOps Practices

Contribution: The integration of Agile and DevOps practices in the testing process has fostered a culture of continuous improvement and collaboration between developers, testers, and operations teams.

Achievement: This approach has reduced the development lifecycle, enhanced product quality, and increased customer satisfaction through iterative updates and immediate feedback.

Implementation of CI/CD Pipelines

Contribution: Continuous Integration/Continuous Deployment (CI/CD) practices have been pivotal in maintaining high-quality standards by integrating testing directly into the software development pipeline.

Achievement: Organizations have been able to deploy updates faster and with fewer errors, enhancing operational efficiency and responsiveness to market changes.

Advancements in Performance and Security Testing

Contribution: Specialized testing frameworks and tools for performance and security testing have been developed, allowing testers to uncover potential vulnerabilities and performance bottlenecks before they impact end-users.

Achievement: Tools like JMeter for performance testing and OWASP ZAP for security testing have become standards in evaluating application robustness and security.

CHAPTER 7

REFLECTION AND ANALYSIS

REFLECTION

1. Software testing has evolved significantly from its early days when it was often an afterthought in the software development lifecycle. Today, it stands as a pivotal phase in the development process, integral to ensuring the delivery of high-quality software products. The integration of methodologies like Agile and DevOps has revolutionized testing by embedding it into every stage of development, promoting a 'test early and often' philosophy. This shift not only enhances product quality but also reduces the time and cost associated with late-stage error corrections.
2. The emergence of automated testing tools has been a game-changer, enabling rapid execution of tests and helping maintain consistency and efficiency. Automation has also facilitated the implementation of continuous integration and continuous deployment (CI/CD) practices, allowing organizations to deliver updates more swiftly and with greater confidence in the stability of their software products.
3. Advancements in artificial intelligence and machine learning are beginning to reshape testing landscapes by predicting potential defects and optimizing testing efforts. Despite these technological advancements, the role of human testers remains critical, particularly in complex scenarios requiring nuanced judgment and understanding of user experience.

ANALYSIS

Software testing has undergone significant transformations, becoming integral to software development for ensuring quality and reliability. The adoption of Agile and DevOps methodologies emphasizes continuous integration and testing, facilitating faster development cycles and frequent releases. This cultural shift underscores the move towards iterative improvements and testing's role in them.

Automation has notably enhanced testing efficiency, enabling faster feedback loops and reducing manual errors. Tools like Selenium and Jenkins support automated testing, vital for maintaining high-quality standards in fast-paced environments..

CHAPTER 8

RECOMMENDATIONS

Continued Learning and Skill Enhancement: Encourage the individual to continue expanding their skill set and staying updated on emerging technologies and industry trends. This could involve pursuing certifications or advanced training programs in areas such as cloud computing, database management, and software development.

Cross-Functional Collaboration: Foster opportunities for the individual to collaborate with colleagues from different departments or teams within Intellipaat. Cross-functional projects can provide exposure to diverse perspectives and approaches, fostering innovation and enhancing problem-solving capabilities.

Leadership Development: Provide opportunities for the individual to take on leadership roles or mentorship responsibilities, allowing them to further develop their leadership skills and inspire others within the organization. This could involve leading project teams, organizing knowledge-sharing sessions, or participating in mentorship programs.

Feedback and Recognition: Offer regular feedback and recognition for the individual's contributions, acknowledging their efforts and achievements. This can boost morale, motivation, and job satisfaction, fostering a positive work environment and encouraging continued growth and performance excellence.

Focus on Impact and Strategic Alignment: Encourage the individual to align their projects and initiatives with the broader strategic goals and priorities of DailyPe. Emphasize the importance of delivering measurable impact and value to the organization, whether through improved efficiency, cost savings, or enhanced customer satisfaction.

CHAPTER 9

CONCLUSION

As I reflect on my journey at Intelipaat and the significant contributions I've made to the organization, I'm struck by the depth of impact that each project has had. The role of software testing within Intellipaat is pivotal to its mission of providing high-quality professional training in various technological domains, including software testing itself. As an educational provider, Intellipaat not only teaches the fundamentals and advanced techniques of software testing but also demonstrates its importance through its own application.

The incorporation of Agile and DevOps methodologies within Intellipaat's development processes underscores the importance of iterative testing and continuous improvement. This approach allows for quicker iterations, faster feedback, and more responsive changes to both content and platform functionalities. Automated testing strategies are essential in this dynamic environment, enabling frequent updates and ensuring that new features are delivered without disruptions or defects.

Moreover, as Intellipaat continues to innovate and expand its technological footprint, the role of software testing becomes increasingly complex and vital. Advanced testing techniques, including performance and security testing, are crucial to ensure that the platform can handle the increasing influx of users and protect their data from potential security breaches. Additionally, the testing team at Intellipaat is tasked with ensuring compatibility across various devices and operating systems, which is critical for delivering a universally positive user experience. A seamless user experience that is fundamental to the learning process. This testing ensures reliability, performance, and security across their services, directly impacting customer satisfaction and trust.

CHAPTER 10

REFERENCES

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